

D624

Biostatistics and Analysis

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A cancer registry, updated each quarter by a team of six abstractors, records tumor stage as I, II, III, or IV. What level of measurement is stage?

- A. Interval, because the stages are numbered.
- B. Ratio.
- C. Ordinal.
- D. Nominal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which variable is measured at the ratio level?

- A. Temperature in degrees Fahrenheit.
- B. Blood type.
- C. Highest education level completed.
- D. Body weight in kilograms.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At a regional hospital that recently added 40 beds, most patients stay 2 to 4 days, but a few stay more than 60. The quality director wants one number for a typical stay. Which should she report?

- A. The median.
- B. The mean.
- C. The mode is always the best choice.
- D. The range.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Systolic blood pressure readings collected at 12 pharmacies over the summer are normally distributed with a mean of 120 mm Hg and a standard deviation of 15. About what percent of readings fall between 90 and 150?

- A. 99.7%
- B. 68%
- C. 47.5%
- D. 95%

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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5. [TRANSFER — reversed stem / different variable]

With the same mean (120) and SD (15), about what percent of readings are above 135?

- A. 16%
- B. 2.5%
- C. 34%
- D. 32%

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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6.

A foundation-funded trial of 2,400 adults finds a walking program lowered blood pressure more than usual care, with $p = 0.03$. Which interpretation is correct?

- A. The effect must be large, because the sample was large.
- B. There is a 3% chance the null hypothesis is true.
- C. If there were no true difference, results this extreme would occur about 3% of the time.
- D. There is a 97% chance the program works.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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7.

A cohort study, published in a well-known journal, reports that night-shift nurses had a relative risk of 1.3 for a condition (95% CI 0.9 to 1.8). What should you conclude?

- A. Night shifts definitely cause the condition.
- B. Not statistically significant at the 0.05 level, because the interval includes 1.
- C. Significant, because 1.3 means a 30% higher risk.
- D. Significant, because the interval excludes 0.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A trial reports a mean difference in weight change of -2.5 kg (95% CI -4.1 to -0.9). What can you conclude?

- A.** The true difference is exactly -2.5 kg.
- B.** Not significant, because the interval excludes 1.
- C.** Not significant, because all values are negative.
- D.** Statistically significant, because the interval excludes 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A small pilot study with 20 participants, run at a clinic that recently changed its name, finds no significant effect of a new smoking-cessation app. A later large trial shows the app truly works. What error did the pilot make?

- A.** Type II error, likely from low power.
- B.** Type I error.
- C.** Selection bias.
- D.** No error; small studies are always right.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Researchers compare mean BMI among adults in urban, suburban, and rural counties. Data came from a survey with a 61% response rate. Which test fits?

- A.** An independent-samples t-test.
- B.** One-way ANOVA.
- C.** Pearson correlation.
- D.** Chi-square test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Across 50 neighborhoods, a walkability score and average adult BMI have $r = -0.6$. The data were mapped by a university GIS lab. Which statement is correct?

- A.** Walkability causes lower BMI.
- B.** About 36% of the variation in BMI is associated with walkability; cause is not shown.
- C.** About 60% of the variation in BMI is explained.
- D.** The relationship is weak, because r is negative.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A regression reports $r^2 = 0.49$ for a single predictor. What could the correlation coefficient r be?

- A.** 0.49.
- B.** 0.7 only, because r^2 is positive.
- C.** 0.7 or -0.7 .
- D.** 0.24.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A cancer registry, ~~updated each quarter by a team of six abstractors,~~ records tumor stage as I, II, III, or IV. What level of measurement is stage?

Target word: level of measurement **Decides:** Ordered categories without equal spacing.

A	Adjacent category	Numbers label order, not equal spacing.
B	Confusion set	No equal gaps or true zero.
C	Correct	Stages are ordered, but the gaps are not equal.
D	Inverted operation	Stages have a clear order.

2. Answer D Concept 1 • transfer

Which variable is measured at the ratio level?

Target word: identify ratio **Decides:** Equal gaps and a true zero.

A	Adjacent category	Interval: zero is not an absence of heat.
B	Inverted operation	Nominal.
C	Confusion set	Ordinal.
D	Correct	Zero means no weight; 80 kg is twice 40 kg.

3. Answer A Concept 2

At a regional hospital ~~that recently added 40 beds,~~ most patients stay 2 to 4 days, but a few stay more than 60. The quality director wants one number for a typical stay. Which should she report?

Target word: best center measure **Decides:** Right skew from long stays.

A	Correct	Resistant to the few very long stays.
B	Inverted operation	Pulled upward by outliers.
C	Absolute language	No measure is best in every case.
D	Confusion set	A spread measure, not a center.

4. Answer D Concept 3

Systolic blood pressure readings ~~collected at 12 pharmacies over the summer~~ are normally distributed with a mean of 120 mm Hg and a standard deviation of 15. About what percent of readings fall between 90 and 150?

Target word: percent within the range **Decides:** 90 to 150 is ± 2 SD.

A	Confusion set	That is ± 3 SD.
B	Adjacent category	That is ± 1 SD (105 to 135).
C	Dropped element	Only one side of the mean.
D	Correct	Two SDs either side of the mean.

5. Answer A Concept 3 · transfer

With the same mean (120) and SD (15), about what percent of readings are above 135?

Target word: upper tail beyond +1 SD **Decides:** $(100\% - 68\%) \div 2$.

A	Correct	32% falls outside ± 1 SD, split evenly: 16%.
B	Confusion set	The tail beyond +2 SD.
C	Adjacent category	The share between the mean and +1 SD.
D	Dropped element	Forgets to split between tails.

6. Answer C Concept 4

A foundation-funded trial of 2,400 adults finds a walking program lowered blood pressure more than usual care, with $p = 0.03$. Which interpretation is correct?

Target word: p-value meaning **Decides:** Probability of results this extreme if no true effect.

A	Camouflage	Large samples can make small effects significant.
B	Confusion set	A p-value is not the probability of the null.
C	Correct	The correct definition of a p-value.
D	Inverted operation	$1 - p$ is not the probability of the alternative.

7. Answer B Concept 5

A cohort study, published in a well-known journal, reports that night-shift nurses had a relative risk of 1.3 for a condition (95% CI 0.9 to 1.8). What should you conclude?

Target word: significance from CI **Decides:** The CI includes 1.

A	Absolute language	Neither significance nor causation is shown.
B	Correct	A ratio CI containing 1 is compatible with no effect.
C	Dropped element	Ignores the interval.
D	Adjacent category	For ratios, the null value is 1, not 0.

8. Answer D Concept 5 · transfer

A trial reports a mean difference in weight change of -2.5 kg (95% CI -4.1 to -0.9). What can you conclude?

Target word: significance of a difference **Decides:** The CI excludes 0.

A	Absolute language	The point estimate is uncertain.
B	Adjacent category	1 is the null for ratios, not differences.
C	Inverted operation	Direction does not decide significance.
D	Correct	For differences, the null value is 0.

9. Answer A Concept 6

A small pilot study with 20 participants, ~~run at a clinic that recently changed its name~~, finds no significant effect of a new smoking-cessation app. A later large trial shows the app truly works. What error did the pilot make?

Target word: error type **Decides:** Missed a real effect.

A	Correct	Failing to detect a true effect.
B	Inverted operation	Type I is a false alarm.
C	Adjacent category	A design problem, not a hypothesis-testing error.
D	Absolute language	Small studies often lack power.

10. Answer B Concept 7

Researchers compare mean BMI among adults in urban, suburban, and rural counties. ~~Data came from a survey with a 61% response rate.~~ Which test fits?

Target word: choose the test **Decides:** One continuous outcome, three groups.

A	Adjacent category	Compares only two groups.
B	Correct	Compares means across three or more groups.
C	Dropped element	Needs two continuous variables; location is a category.
D	Confusion set	For categorical outcomes, not means.

11. Answer B Concept 8

Across 50 neighborhoods, a walkability score and average adult BMI have $r = -0.6$. ~~The data were mapped by a university GIS lab.~~ Which statement is correct?

Target word: interpret r **Decides:** $r^2 = 0.36$; correlation only.

A	Adjacent category	Correlation does not establish cause.
B	Correct	$r^2 = 0.36$, and correlation does not prove causation.
C	Dropped element	Forgets to square r .
D	Inverted operation	Sign shows direction; size shows strength.

12. Answer C Concept 8 • transfer

A regression reports $r^2 = 0.49$ for a single predictor. What could the correlation coefficient r be?

Target word: solve for r **Decides:** $r = \pm\sqrt{0.49}$.

A	Wrong stem number	That is r^2 , not r .
B	Dropped element	Ignores that r may be negative.
C	Correct	The square root of 0.49 is 0.7, and r^2 hides the sign.
D	Inverted operation	Squares 0.49 instead of taking the root.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Levels of measurement	Nominal = unordered categories; ordinal = ordered categories with uneven gaps; interval = equal gaps, no true zero; ratio = equal gaps and a true zero.	Calling any numbered category interval, or treating Fahrenheit as ratio.	Whether categories have order, equal spacing, and a meaningful zero.
2	Mean vs. median in skewed data	In right-skewed data a few high values pull the mean above the median, so the median better describes the typical value.	Reporting the mean by habit when outliers are present.	Most values bunched low with a few very high ones.
3	The empirical rule	In a normal distribution, about 68% of values fall within 1 SD of the mean, 95% within 2, and 99.7% within 3.	Forgetting to split the outside share between the two tails.	A mean, an SD, and a range or cutoff.
4	What a p-value means	A p-value is the probability of results at least this extreme if the null hypothesis were true; it is not the probability that the null is true.	Reading $p = 0.03$ as a 3% chance the null is true or a 97% chance the program works.	A reported p-value and several interpretations.
5	Confidence intervals and significance	A 95% CI for a ratio that includes 1, or for a difference that includes 0, is not statistically significant at the 0.05 level.	Declaring an effect significant because the point estimate looks large.	A ratio or difference reported with its 95% CI.
6	Type I and Type II errors	A Type I error rejects a true null (a false alarm); a Type II error misses a real effect, which small samples and low power make more likely.	Swapping the two error types.	A study conclusion compared with what is actually true.
7	Choosing a statistical test	Chi-square compares categorical variables; a t-test compares two group means; ANOVA compares three or more means; correlation or regression relates continuous variables.	Running multiple t-tests or a chi-square on means.	The number of groups and whether the outcome is a category or a measurement.
8	Correlation and r^2	r shows strength and direction of a linear link; r^2 is the share of variation explained; neither proves causation.	Reading r as the percent explained or as proof of cause.	A correlation coefficient and a claim about cause or variance explained.

Course scope (public description): Biostatistics and Analysis introduces learners to the role that biostatistics has in public health, predicting public health outcomes and influencing decisions made with policies. Learners will examine the sources and methods of collecting public health information, analyze and interpret quantitative data in research studies, and make inferences on the effectiveness of health program interventions and policies. There are no prerequisites for this course.